

JORDI CORBILLA

SENIOR/PRINCIPAL/LEAD SOFTWARE ENGINEER AND LEAD ARCHITECT | BRITISH CITIZEN RESIDING IN LONDON

As a Senior Engineer, I possess a strong and diverse set of skills that allow me to effectively manage multiple projects and meet strict deadlines, while also understanding complex and interdependent business processes and products. With over 20 years of experience in Software Engineering, including 12 years as a Lead Engineer across various industries such as Hedge Funds, Fintech, and Industrial Automation, I have a proven track record of success in delivering high-quality software solutions using cutting-edge technologies such as .NET, Python, React, and AWS.

I am skilled at identifying client needs and translating them into appropriate technical solutions, and I am dedicated to fostering a culture of excellence and collaboration by effectively communicating with all organizational levels and cross-functional teams. I have a strong background in mentoring and training, and I am adept at inspiring and motivating others to embrace new challenges and work together to achieve shared goals.

As a natural leader, I believe in gaining buy-in and building mutual respect by communicating openly, encouraging feedback, and providing a supportive environment for growth and development. My passion for technology and dedication to driving results make me an asset to any team or organization.

EXPERTISE

- | | | |
|----------------------------|----------------------------|-----------------------------|
| • Management & Leadership | • TDD, BDD, DevOps & SOLID | • Continuous Improvement |
| • Talent Development | • Project Management | • Change Management |
| • Compliance | • Trade capture | • Matching, Settlement |
| • Order Management System | • Mentoring & Coaching | • Problem Solving |
| • Financial Tech Solutions | • Solutions Architecture | • UI/UX design |
| • Hedge Funds | • Microservices & Cloud | • Negotiating & influencing |
| • Optimization | • Machine Learning & AI | • Code Reviews |

EXPERIENCE

Balyasny Asset Management – London (UK)

Sep-2020 - present

Senior Software Engineer, Trading Execution & Compliance Technology – BAM

- Designs, builds, and extends trading and compliance technology platforms at BAM using cloud services, containers, REST API, and React.
- Works closely with trading, compliance, and technology teams to deliver cutting-edge innovative software solutions that comply with diverse industry regulations and requirements, enabling BAM to enter new markets with increasingly complicated regulatory needs.
- Proficient in Markets in Financial Instrument Directive/Regulation (MiFID II) Trade Regulatory Reporting for T0 and T1, ESMA/FCA Short Selling for Europe/UK, Short Position Reporting, Market Data/Pricing APIs via BPIPE, and Monetary Authority of Singapore (MAS) OTC Regulatory Reporting.
- Skilled in designing, scaling, optimizing, and refactoring large-scale systems with diverse data locality, as well as in pre-trade surveillance and research process automation, trade execution rules and exposures automation, ownership, short reporting, post-trade allocation, and regulatory filings' automation.
- Experienced in market abuse automation, restrictions, and exclusions.

Notable Achievements:

- Automated several complex manual systems to reduce the operational risk for the Compliance & Trading Teams.
- MiFID 2 regulatory improvements.
- Implemented MAS OTC Reporting for Equities, FX, and Commodities.
- Built the internal Compliance Filing tool and the Expert Network tool using .NET 6 and React.
- Introduced Spec Flow testing as part of the micro-services approach.
- Led cloud adoption using AWS, Harbor, Rancher, Helm, Python, .NET 6, React, Docker, and Kubernetes.
- Mentored internship fellows as part of the [BAM ATLAS program](#).

VR Capital Group – London (UK)

Feb-2016 – Sep-2020

Director of Application Development – VR Capital

Feb-2019 – Sep-2020

- Led end-to-end enterprise solutions and provided full digital transformations for new products and services, taking full ownership of deliveries, technical roadmaps, choice of technologies, best practices, KPI measurement, quality, and timelines.
- Approached projects with a forward-thinking mindset and utilized disruptive technologies.
- Improved route-to-live processes through the implementation of DevOps, testing automation, and quality control.
- Directed and coached distributed agile teams, ensuring accountable incremental transformation daily to increase the value delivered through technology to business stakeholders.
- Optimized performance and built robust intelligent solutions to problems, while also bringing in new ideas and innovations.
- Architected and developed critical-mission finance/trading systems using C#, Python, ASP.NET core, SQL Server, TDD, Docker, Kubernetes, responsive websites (HTML5), React, and Blazor.

Notable Achievements

- Led the OMS system project on a 4 year-plan to add 20 different asset types (FX, Repo, FI, Equities, Futures, Swaps, CDS/CDX, OTC, etc.) including P&L interaction, Risk and Compliance capabilities, supporting thousands of trades per day and web/mobile availability.
- Led all internal financial products and total reward systems (bonus scheme, salaries, trading limits).
- Led a team of developers and testers based in Ukraine with a \$1m/year budget.
- Led the infrastructure migration to achieve high availability.
- Led the Agile transformation and DevOps culture adoption.

Lead Software Engineer / Solutions Architect – VR Capital**Feb-2016 – Feb-2019**

- Architected and developed in-house investment, trading, and analytics systems using a full-stack of technologies including ASP.NET, WPF, Rx, PHP, C#, SQL Server, TDD, MVC, JavaScript, Python, CSS, jQuery, SignalR, Knockoutjs, Angular, Reactjs, XML, API, Web Services, HTML, DHTML, Design Patterns, and REST.
- Provided support to trading desks across multiple locations, including New York, Argentina, Chicago, Moscow, Dubai, and London.
- Led the design and specification of all system components, taking into consideration functionality, data, security, integration, infrastructure, and performance.
- Handled multiple projects while assessing other developer's workloads.
- Contributed to the platform redesign and architectural choices to shape the future of technology for the group.
- Collaborated with remote development teams located in New York and Ukraine to build a global team.
- Helped establish a long-lasting and repeatable software development standard across the team using Agile methodologies.
- Participated in code reviews across all areas of the team to improve the quality of the delivered solutions.
- Contributed to the overall security and vulnerabilities of the applications following OWASP standards.
- Established a culture of TDD, BDD, and CD.
- Worked across green-field projects to support new market opportunities, improve process efficiencies, reduce operational risk, improve platform scalability, and ensure regulatory compliance.
- Collaborated with trading desks worldwide to deliver strategic end objectives.
- Expertise in MIFID II Trade Regulatory Reporting, RHUB, and FIX Messaging.
- Implemented Trade Surveillance to store all messaging/email and phone conversations as part of compliance regulations.
- Supported legacy ASP.NET and PHP applications.
- Delivered high-quality IT service within the company (network, phone system, security, backups, etc.).
- Worked with experienced .NET C# developers to help design, code and test a wide range of key business software applications utilized throughout Front, Middle, and Back Office business areas.

Notable Achievements

- Led the company-wide transformation towards a culture of continuous delivery and innovation, achieving higher agility and responsiveness to market demands.
- Collaborated with Traders and PMs to automate various aspects of their work, including FX roll-overs, model automation, ownership alerts, and shorts filing, resulting in increased efficiency and accuracy.
- Developed an internal architecture that facilitated the onboarding of multiple teams and the early adoption of DevOps, improving collaboration and streamlining the software development process.
- Recognized for subject matter expertise and success, obtained a directorship role at VR Capital.
- Successfully delivered a 70% reduction in time across departments, minimizing human error in data entry and increasing overall efficiency and productivity.

Broadridge Financial Solutions Ltd – London (UK)**Mar-2011 – Feb-2016****Lead Technical Architect - Broadridge**

- Led the enhancement of current solution [PROactive™ Confirmation Matching](#) and developed end-to-end software solutions in an agile environment.
- Possesses business process knowledge including trade matching, HFT, settlement, and SWIFT.
- Designed scalable systems with architectural requirements for high availability, high throughput, and low response time.
- Proficient in Object-Oriented Analysis and Design with appropriate use of design patterns.
- Handled huge data and processed large amounts of information providing high response systems under SQL Server 2008 R2/2012/14.
- Expert in unit testing with DUnit, code coverage, and continuous integration using TeamCity.
- Continuously improved system architecture by refactoring old legacy code.
- Designed systems or components on large projects and modelled mathematic algorithms to display performance rates for multi concurrent entities.
- Proficient in source control management using TFS, Subversion, and GitHub.
- Possesses architecture domain skills (Data, Application, and Infrastructure) using a good understanding of architectural concepts, methodologies, and mapping approaches.
- Led project management of new products, making key technical decisions, and guiding and mentoring the team around technical challenges.
- Worked with outsourced or offshore partners to deliver software projects.
- Collaborated with other parts of the technology team, including product, QA, etc. to deliver solid products.
- Experienced in testing methodologies, test-driven design, and the creation of effective and reliable test cases.
- Advocated the use of the Domain Driven Design, Inversion of Control, good Object-oriented design practices, MVC, Single Responsibility Principle, Gang of Four Design Patterns, Open/Closed Principle, Simplicity of Design, UML, TDD, Code Coverage, BDD, and other similar practices that help create solid, maintainable code.
- Conducted code review providing feedback and mentored/trained people in India to increase their technical ability.
- Reviewed technical requirements provided by the business for key projects and provided solid software/component level designs supporting technical requirements.
- Brought teams to a higher level of performance and is highly efficient in Agile processes.
- Expert in Continuous Integration, Continuous Deployment, and Continuous Delivery and used them in several projects across the company.
- Developed automation testing suite for cross-browser testing using Ruby, Watir, and Browser Stack.
- Configured and set up Build Agents infrastructure to perform Continuous deployment using VMWare machines with Web Deploy and TeamCity.

Notable Achievements

- Architected and developed the [PROactive™ Matching High Frequency Trading](#) solution:
 - Designed and developed a state-of-the-art HFT solution that automates the processing of trade confirmations, enabling the system to cope with algorithmic trading and HFT (High frequency trading) at a rate of 1 million trades/day, where the engine was able to process 36 trades a second or 2160 trades a minute.
 - Led the development process for 24 months and implemented the solution in Bank of America New York, using Windows platform and concurrent modules.
 - Used Delphi and .NET technologies over SQL Server and driven the development process through agile programming methods.
 - Achieved unbeatable match rates and exception workflow from the earliest point in the pre-settlement cycle, making this solution the market leading confirmation matching system in the Treasury Management System space.
- Architected and developed the [PROactive™ eMatching](#) solution:
 - Developed a web portal for internal banking and corporates, called PROactive™ eMatching, using ASP.NET, MVC, JavaScript, CSS, jQuery, SQL Server and other .NET technologies.
 - Supervised a team of 8 developers (4 in the UK and 4 in India) in the development process for 24 months.
 - The solution is currently live at Commerz Bank, handling thousands of users and thousands of requests per second.

[AEtech - Girona \(Spain\)](#)

Jun-2003 – Mar-2011

Senior Software Engineer – AEtech

Aug-2007 – Mar-2011

As a senior software engineer, I was responsible for project architecture, conceptual and functional design, development, and deployment, ensuring that the "big picture" was always considered while ensuring that the "small details" were properly addressed in the implementation. I have extensive experience in full DB integration, design, and performance optimization with SQL Server 2000 - 2005 backend. I also proposed and managed the implementation of proper DB versioning and deployment procedures.

My organizational and prioritization skills helped me to complete tasks efficiently and achieve desired results. I designed and implemented necessary deployment processes related to applications managed, developed necessary automated scripts, and created and maintained required support documentation. My experience with third-party VCL components (DevExpress), knowledge of Win32 API, TCP/IP protocols, and socket programming, along with the ability to design and develop in OOAD - Object-Oriented Analysis/Design (UML, Design Patterns), made me proficient in developing robust software systems. I also have experience in database management using SQL and UML.

Notable Achievements

- Designed and managed the implementation of the company's front-end clients (Win32 with Delphi 2007-XE) and core Application Server modules, including infrastructure, data feeds, multithreading data processing and analysis, data manipulation, and aggregation.
- Developed internal framework scheduling system solution (AEmes Scheduling) in collaboration with UPC (Polytechnic University of Catalonia) to increase productivity in batch factories. The code is exclusively Delphi with widespread use of VCL for the GUI front-end.
- Developed a timing sequencing algorithm for cluster tasks based on the following paper "[Design and development of a timing sequencing algorithm for cluster tasks](#)".
- Developed an internal open-source framework ([Zaluum](#)) for industrial automation based on Java in collaboration with UPC. I set up the multithreading system on an RTOS (Real-time OS) with Ubuntu Server and JVM RT (Real-time Java VM).
- Developed [AEmes](#) (Manufacturing Execution System) ERP product server-side/client-side. Created AI algorithms to achieve high performance with low latency.
- Developed a library in C# to automate the drawing of schemes in AutoCAD 2009 using COM and Delphi.

Senior Automation Control Engineer – AEtech

Jun-2003 – Aug-2007

As a senior automation control engineer, I possessed strong technical knowledge in PLC architecture and programming, HMI/SCADA programming, and DCS systems. I was responsible for drive commissioning, soft starters commissioning, and PID design, technical documentation, monitored work for large customers, support, and technical service. I provided technical support for production operations, including troubleshooting equipment and process problems. I actively participated in all phases of assigned projects, including design, fabrication, off-line integration and testing, installation, production start-up support, training of plant personnel, and transfer to operations. I have expertise in a full array of automation technologies, including PLC and HMI programming, panel design, and system integrations. I have hands-on knowledge of electrical installation, am a self-starter, an independent and quick learner, and have experience in data retrieval, plotting, and analysis using Microsoft Excel. I am experienced with MES application development and familiar with Minitab and statistical process control.

Notable Achievements

- Created simulation models for production improvements using Matlab.
- Developed new or improved existing product designs to address critical quality requirements.
- Developed methods, measurements and processes to improve capabilities related to manufacturing.
- Improved hundreds of manufacturing processes and products.

EDUCATION & PERSONAL DEVELOPMENT

MSc in Computer Engineering, Open University of Catalonia (Spain), 2014 - 2019

- Focused on Advanced AI, Machine Learning, and Distributed Systems.
- Conducted Master's thesis on "[Ocular disease intelligent recognition through Deep learning architectures](#)."

BSc (Hons) Computer Engineering, Open University of Catalonia (Spain), 2006 - 2014

- Emphasis on Grammars, Compilers, Knowledge representation, Data mining, computerised learning and AI.
- Conducted Bachelor's thesis on "[Distributed decision-making system](#)."

BEng (Hons) Industrial Electronics Engineering, University of Girona (Spain), 1999 - 2003

- Emphasis on Electronics, Robotics, and Industrial Automation.
- Conducted Bachelor's thesis on "[Monitoring of a remote laboratory via Data Socket](#)."

Latest certification and training:

- 2023 Build Microservices with .NET & Amazon Web Services, Udemy.
- 2023 .NET Microservices: CQRS & Event Sourcing with Kafka, Udemy.
- 2023 Deploy Machine Learning Models on AWS Lambda, Udemy.
- 2022 Machine Learning Applied to Stock & Crypto Trading, Udemy.
- 2022 Statistical Arbitrage Bot build in Crypto with Python, Udemy.
- 2022 Python for Data Science and Machine Learning. Udemy.
- 2022 Learn to build an e-commerce store with .NET 6, React & Redux, Udemy.
- 2022 Introduction to Unity for 2D Video Games (Ongoing). Domestika.
- 2021 AWS Solutions Architect Certification (Ongoing), Udemy.
- 2021 Crypto Trading with QuantConnect, Udemy.
- 2021 The Complete Cryptocurrency Investment course, Udemy.
- 2021 Data Science: Natural Language Processing (NLP) in Python, Udemy.
- 2021 Stock Trading: Invest and Trade Safely with Stock Sector ETF, Udemy.
- 2021 React – The Complete Guide (Hooks, React Router, Redux), Udemy.
- 2020, ASP.NET Core API (.NET 5.0) and Angular 10, Udemy.
- 2020, Algorithmic Trading & Quantitative Analysis Using Python, Udemy.
- 2020, The Complete Foundation Stock Trading Course, Udemy.

TECHNICAL EXPERTISE

• Data Storage: SQL Server 2012-2019, MySQL, Oracle • FIX Systems: QuickFixN, FIXEdge • Languages: C#, Python, Delphi, Java, C++, C, VB6, Ruby, R, Lua, Powershell • Web technologies: ASP.NET Webforms, ASP.NET MVC 5, Web API's (REST, json), jQuery, JavaScript, CSS, Ajax, HTML 5, ASP.NET Core 3.1/5/6, SignalR, Knockoutjs, Reactjs, Angular 10, Blazor, Swagger. • Desktop: Winforms, WFP, VCL, FMX, .NET 4.8, .NET Core 3.1, .NET 5/6	• OMS / EMS Applications: Bloomberg, FlexTrade, Traiana, MS Passport • Middle Office: Geneva and Arcesium • Technology: TCP/IP, COM & DCOM, ActiveX, WCF, Entity Framework, Multi-threading, HPC, AI and ML • CI/CD: TeamCity, Octopus, Jenkins, Prometheus and Grafana • Scheduling: Active Batch • Mobile: Android Studio, PhoneGap • Logging: Serilog, Log4net • Components: TelerikUI, DevExpress • Security: OWASP, SSL, HTTPS, IIS Configuration, Pen Testing	• Regulatory Reporting MiFID II/MAS OTC: TradEcho, RHUB, Unavista, SS&C, DTCC. • Source control: TFS, Git, Svn • Messaging: IBM WebSphere, MSMQ, Kafka, RabbitMQ. • Cloud: Azure, AWS (S3, CloudFront). • Agile Applications: JIRA, Confluence, Redmine and TFS • Containers: Docker and Kubernetes • Deep Learning: TensorFlow and Keras • Reporting: SSRS • Testing: SpecFlow, MSTest, Nunit • Data Analysis: Python (NumPy, SciPy and Pandas)
--	--	--

ADDITIONAL INFORMATION

- **Nationality:** British/Spanish
- **Languages:** Spanish (Native), Catalan (Native), English (Native), Portuguese (Fluent)
- **Interests:** Programming and Photography
- **Awards:** [Embarcadero MVP](#)
- **Blog:** [Random thoughts on coding and technology](#)
- **GitHub:** <https://github.com/JordiCorbilla>
- **LinkedIn:** <https://www.linkedin.com/in/jordicollcorbilla/>